

701 NODE JS • PRACTICAL ASSIGNMENT 3

Practical Submission & Output Report

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Division Details:	Division: B	Semester / Class:	SEM 7 • B.Sc IT
Subject:	701 NODE JS	Design Theme:	Black & Blue Combination

QUESTION 1

Express.js • EJS • Multer • Express-Validator

User Registration Form with File Upload & Validation

Requirement: Develop a user registration form (fields: username, password, confirm password, email, gender, hobbies, upload profile pic, upload other pics) (use text box, radio, checkboxes, file upload) using Express. Form should also contain file upload (single, multiple) with validations. In case of invalid data, display errors along with all previous field values. In case of all valid fields, display all data and images in a well formatted tabular form and allow the user to download that file using Express. Develop route for file download. (Use express, ejs, express-validator)

User Registration Form

Username:

naitik_vaghasiya

Password:

Confirm Password:

Email:

naitik@university.edu

Gender:

☒ Male

☐ Female

Hobbies:

☒ Reading

☐ Music

☒ Sports

Upload Profile Picture (Single):

Choose File

No file chosen

Upload Other Pictures (Multiple):

Choose Files

No file chosen

Register

Figure 1.1: User Registration Form with Fields, Radio, Checkboxes & File Uploads

Registration Details

Field	Value
Username	naitik_vaghasiya
Email	naitik@university.edu
Gender	Male
Hobbies	Reading, Sports
Profile Picture	 Download Profile Picture
Other Pictures	 Download File  Download File

[← Back to Registration](#)

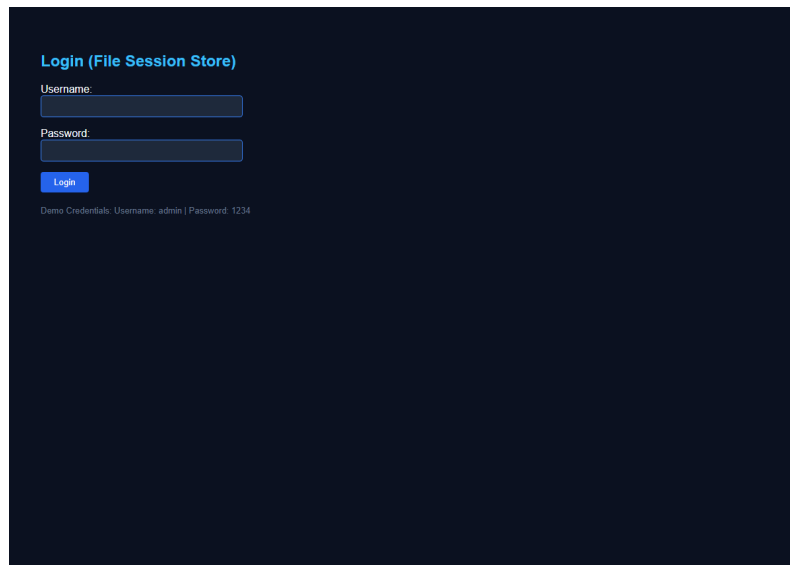
Figure 1.2: Validated Submission Record Table & File Download Links

QUESTION 2

Express.js • EJS • session-file-store

Express Login Application with File Session Store

Requirement: Express Login application with file session store. Functionalities are login, 2 protected routes, Logout.



Login (File Session Store)

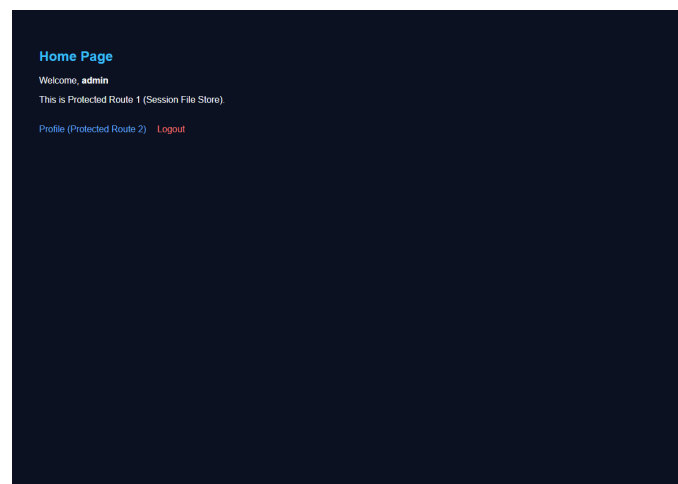
Username:

Password:

Login

Demo Credentials: Username: admin | Password: 1234

Figure 2.1: Login Page with File Session Store (Username: admin | Password: 1234)



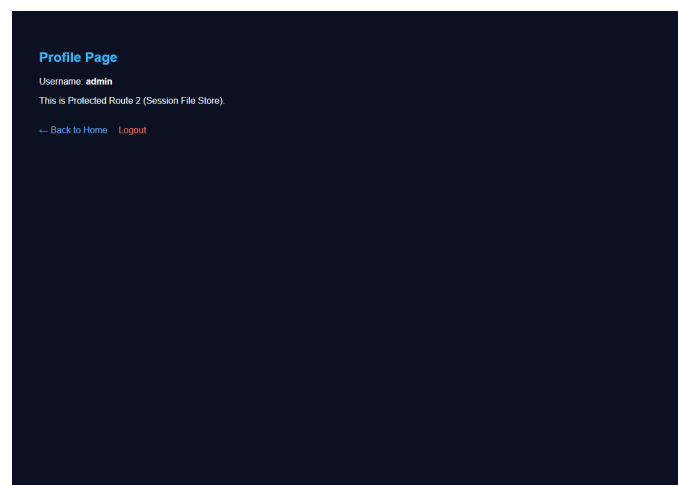
Home Page

Welcome, admin

This is Protected Route 1 (Session File Store)

Profile (Protected Route 2) Logout

Figure 2.2: Protected Route 1 (Home Page)



Profile Page

Username: admin

This is Protected Route 2 (Session File Store)

Back to Home Logout

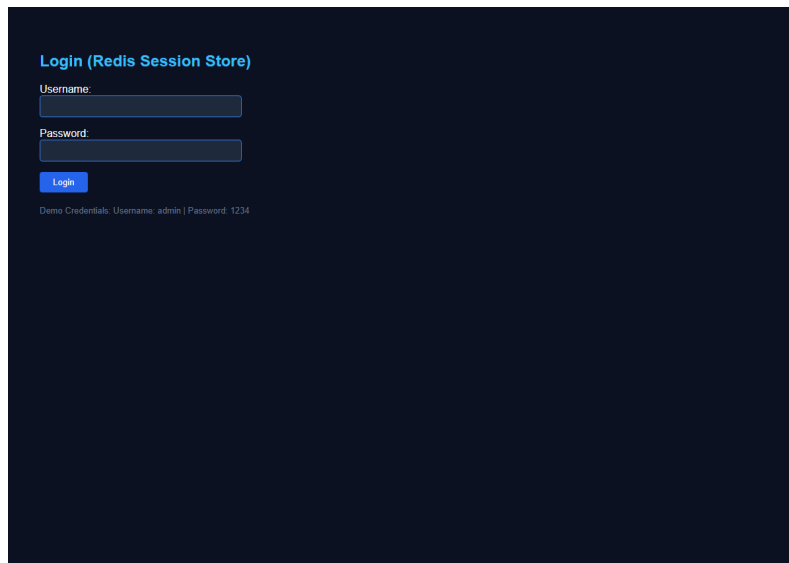
Figure 2.3: Protected Route 2 (Profile Page) & Logout

QUESTION 3

Express.js • EJS • connect-redis • Redis Store

Express Login Application with Redis Session Store

Requirement: Express Login application with redis session store. Functionalities are login, 2 protected routes, Logout.



Login (Redis Session Store)

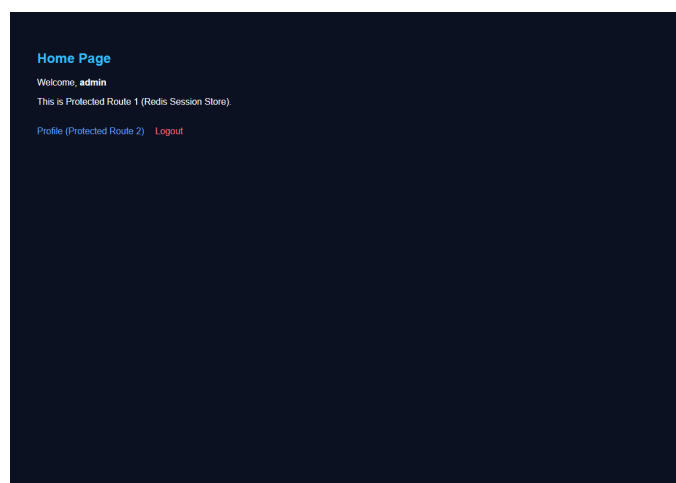
Username:

Password:

[Login](#)

Demo Credentials: Username: admin | Password: 1234

Figure 3.1: Login Page with Redis Session Store (Username: admin | Password: 1234)



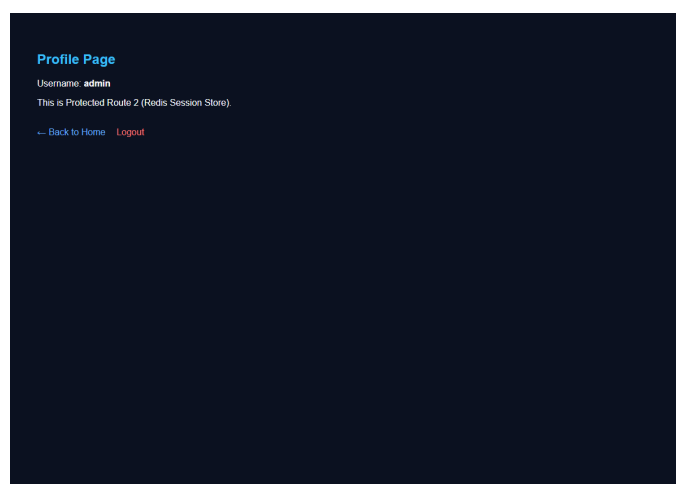
Home Page

Welcome, admin

This is Protected Route 1 (Redis Session Store).

[Profile \(Protected Route 2\)](#) [Logout](#)

Figure 3.2: Protected Route 1 (Redis Session Store)



Profile Page

Username: admin

This is Protected Route 2 (Redis Session Store).

[Back to Home](#) [Logout](#)

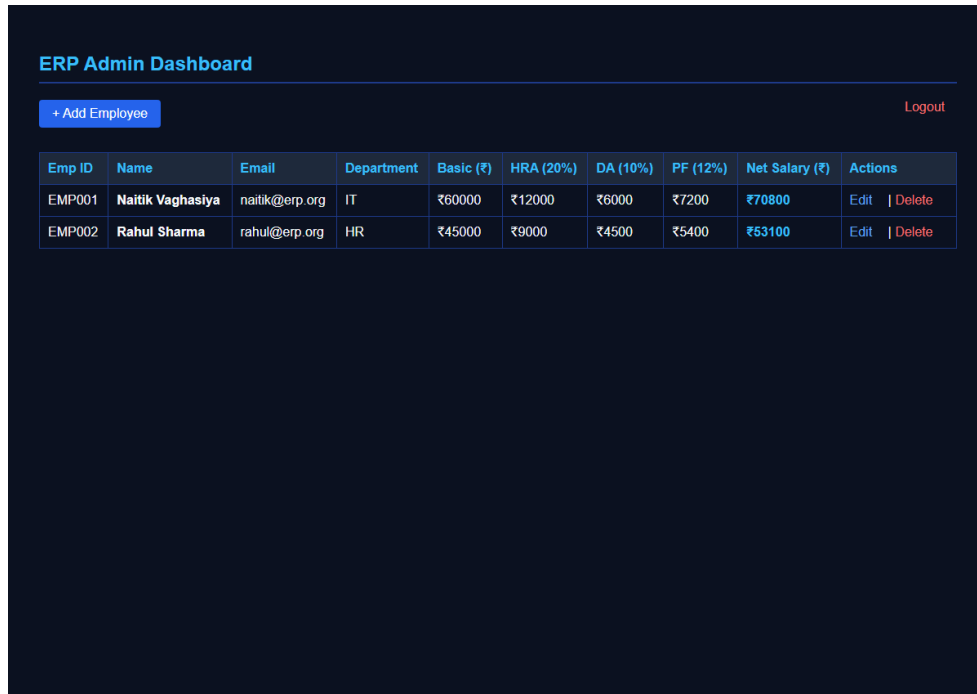
Figure 3.3: Protected Route 2 (Profile Page) & Logout

QUESTION 4

Express.js • EJS • Mongoose • Bcrypt • Nodemailer

ERP Admin Panel with Session, CRUD, Salary Calc & Email

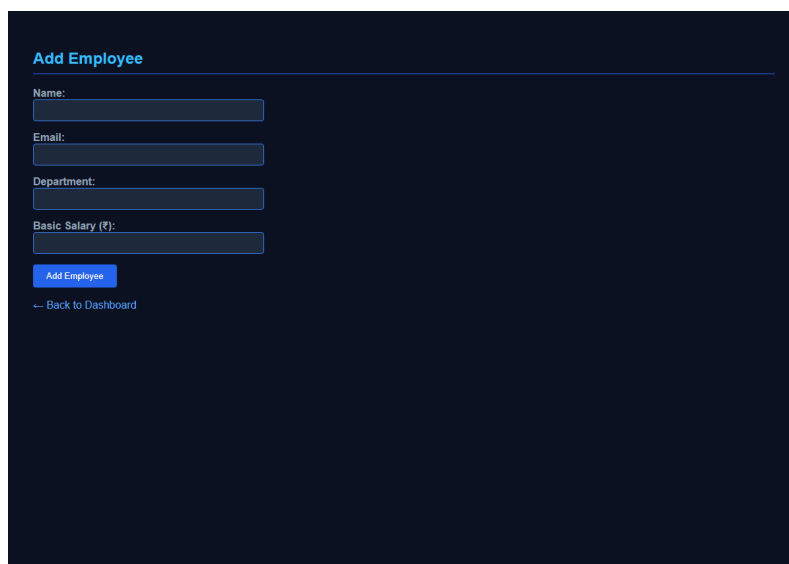
Requirement: Develop login for an ADMIN panel of ERP system with session, CRUD operations for employees table (generate empid and password) with salary calculation with mongoose, express and EJS template engine. Send email to employee when inserted. Encrypt the password. Logout.



The screenshot shows the 'ERP Admin Dashboard' with a '+ Add Employee' button and a 'Logout' link. Below is a table listing two employees:

Emp ID	Name	Email	Department	Basic (₹)	HRA (20%)	DA (10%)	PF (12%)	Net Salary (₹)	Actions
EMP001	Naitik Vaghasiya	naitik@erp.org	IT	₹60000	₹12000	₹6000	₹7200	₹70800	Edit Delete
EMP002	Rahul Sharma	rahul@erp.org	HR	₹45000	₹9000	₹4500	₹5400	₹53100	Edit Delete

Figure 4.1: ERP Admin Dashboard with Employee CRUD & Computed Salary (HRA, DA, PF, Net Pay)



The screenshot shows the 'Add Employee' form with the following fields and buttons:

- Name:
- Email:
- Department:
- Basic Salary (₹):
-
- [← Back to Dashboard](#)

Figure 4.2: Add Employee Form with Auto-Generated EmpID, Encrypted Password & Email Dispatch

QUESTION 5

React.js • Express.js • JWT • Mongoose

Employee Site Login with JWT, Profile & Leave Application

Requirement: Develop employee site (refer Q4) login with JWT, for employees with mongoose, express and React.js. Create a home page with following links: - Page 1) display employee profile. - Page 2) Application for leave (fields date, reason, grant yes/no) Add, List - Logout.

Employee Profile (JWT Authenticated)

Welcome, Employee ID: **EMP087**

Emp ID	EMP087
Name	Naitik Vaghasiya
Email	naitik@erp.org
Department	IT Engineering
Basic Salary	₹,160,000
HRA (20%)	₹,12,000
DA (10%)	₹,6,000
PF (12%)	₹,17,200
Net Salary	₹,70,800

Leave Application

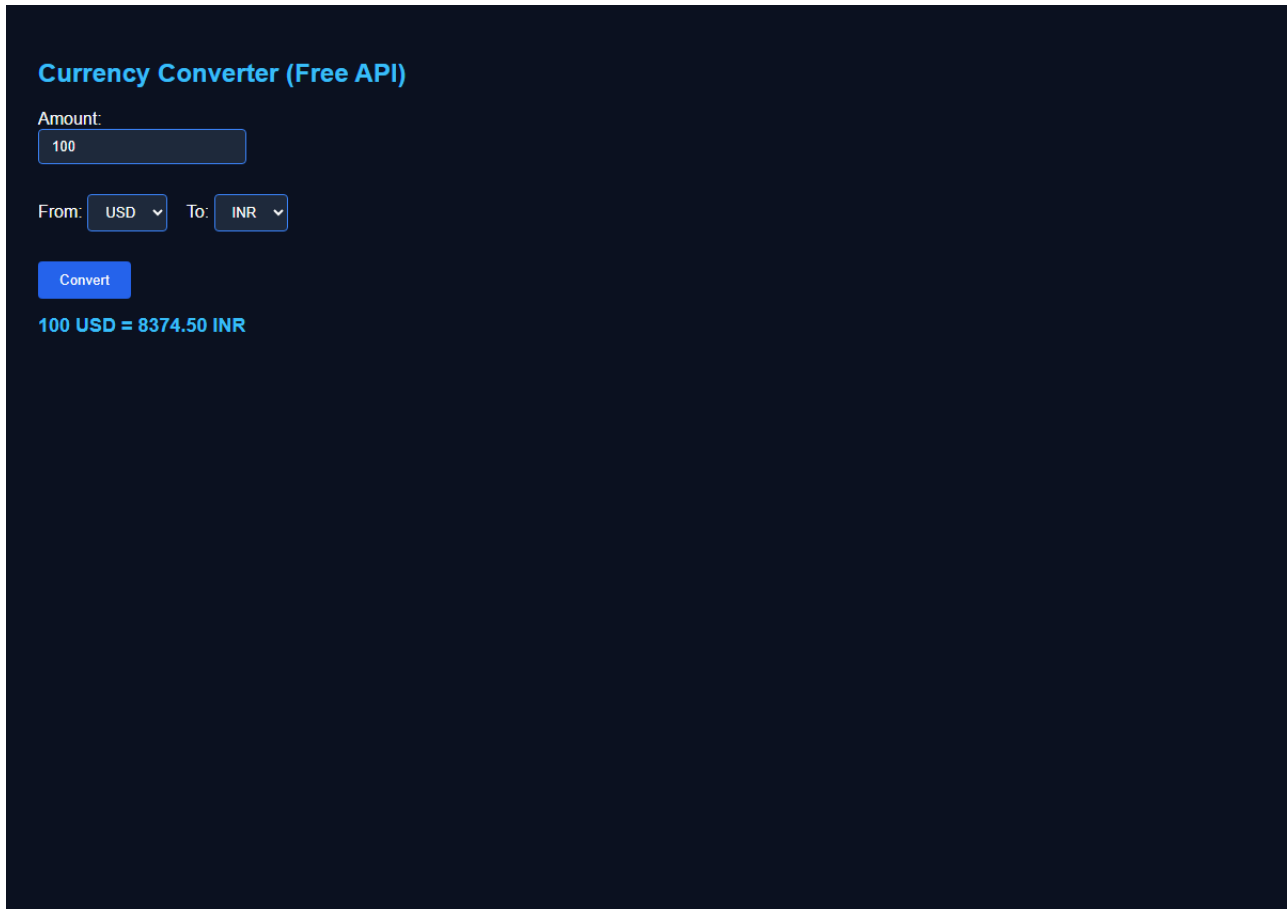
Date: 2026-10-15 | Reason: Medical Checkup | Grant: Yes

Date	Reason	Grant Status
2026-10-15	Medical Checkup	Yes
2026-09-20	Academic Seminar	Yes

Figure 5.1: Employee Site - Page 1 Profile Display & Page 2 Leave Application (Add & List)

QUESTION 6

React.js • Express.js • Free FX Utility API

Free API Utility Integration from Frontend and Backend*Requirement: Call any free api for a useful utility from frontend and backend.*

The screenshot displays a web application titled "Currency Converter (Free API)". It features a form with the following elements: a label "Amount:" followed by a text input field containing "100"; a "From:" label with a dropdown menu set to "USD"; a "To:" label with a dropdown menu set to "INR"; and a blue "Convert" button. Below the form, the result is displayed as "100 USD = 8374.50 INR".

Figure 6.1: Currency Converter Utility calling Free API from Frontend via Backend

QUESTION 7

MERN Stack (MongoDB, Express, React, Node.js)

MERN Shopping Cart with 2-Level Category & Products

Requirement: Shopping cart with admin site and user site with (2 level) Category and Products using MERN stack.

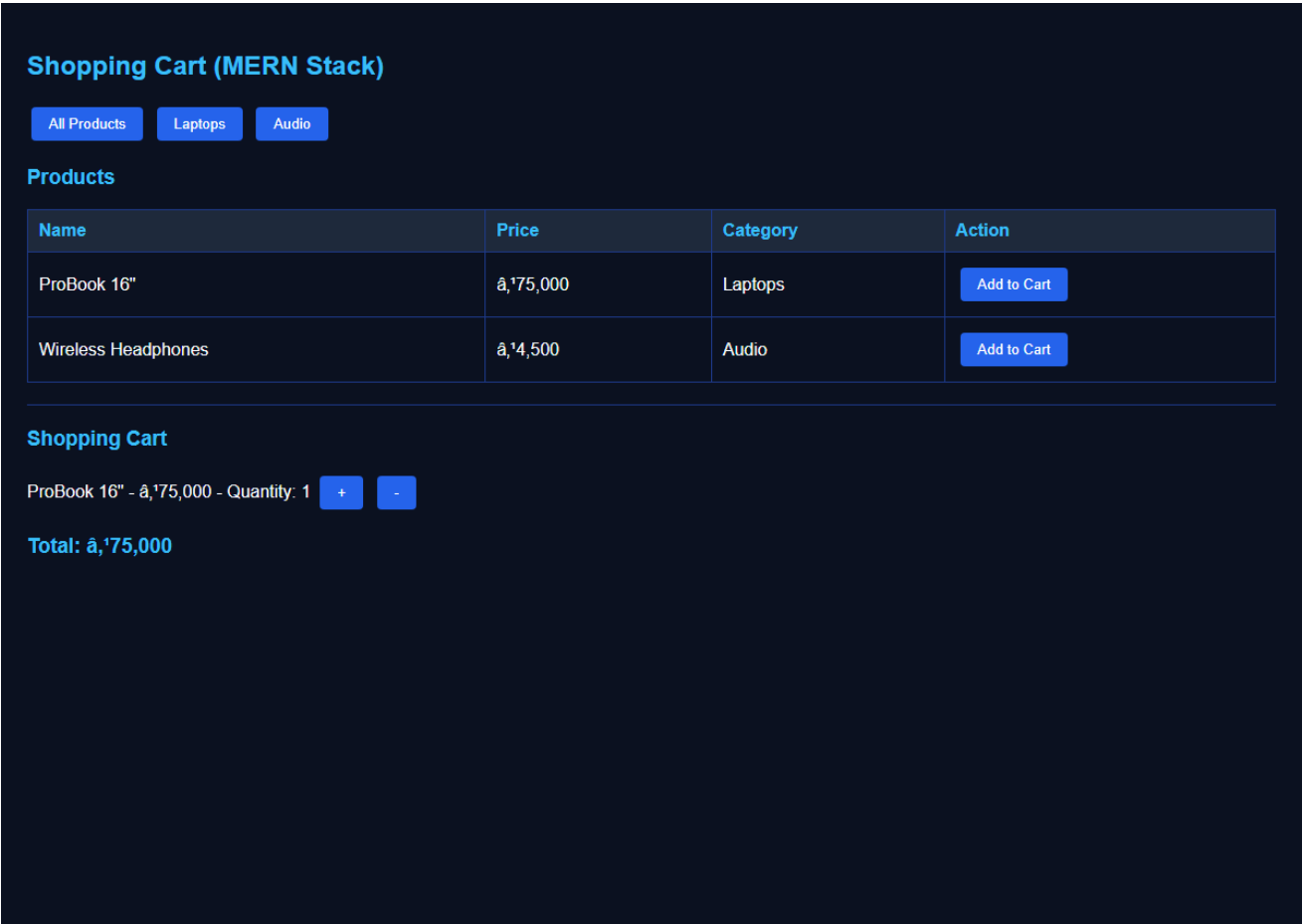


Figure 7.1: MERN Shopping Cart - Products by Category, Add to Cart & Live Total Calculation

QUESTION 8

React.js • Express.js • Sequelize ORM

Student Collection CRUD using Sequelize ORM & React

Requirement: Crud operation for Student collection using Sequelize. (Express + Sequelize + React)

Student Information System (Sequelize CRUD)

Add Student

Add Student

Student List

ID	Name	Email	Age	Course	Actions
#87	Naitik Vaghasiya	naitik@university.edu	21	B.Sc. IT (Sem 7)	<button>Edit</button> <button>Delete</button>
#88	Aarav Patel	aarav@university.edu	22	M.Sc. IT	<button>Edit</button> <button>Delete</button>

Figure 8.1: Student CRUD Operations with Sequelize ORM (Add Student Form & Student Directory Table)